

# Lecture 12: Bayesian analysis

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# Outline

1 Bayesian linear regression

2 Bayesian model selection

# Prior of $\beta$ and $\sigma^2$

In normal linear regression, for  $\{(y_i, x_i^T)\}_{i=1}^n$ ,  $x_i \in R^p$ , we assume

$$y|X, \beta, \sigma^2 \sim MVN(X\beta, \sigma^2 I_n).$$

where  $y = (y_1, \dots, y_n)^T$ , and  $X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix}$ .

- To establish Bayesian model, we assume prior distribution for  $\beta$  and  $\sigma^2$ :

$$(\beta, \sigma^2) \sim \pi(\beta, \sigma^2)$$

Then posterior of  $\beta$  and  $\sigma^2$  are

$$\pi(\beta, \sigma^2) \propto P(y|X, \beta, \sigma^2)\pi(\beta, \sigma^2)$$

- If posterior distribution is exact, then

$$\tilde{\beta} = E[\beta|y, X], \quad \tilde{\sigma}^2 = E[\sigma^2|y, X]$$

which are the posterior estimates of  $(\beta, \sigma^2)$  in Bayesian linear regression.

# Semi-conjugate prior

## Definition (Semi-conjugate prior)

Suppose  $\theta = (\theta_1, \theta_2)^T$ , the semi-conjugate prior is a product of priors,

$$\pi(\theta) = \pi(\theta_1)\pi(\theta_2)$$

each of which, conditioned on the other parameters, is conjugate to  $p(y|\theta)$ .

The Semi-conjugate prior for  $(\beta, \sigma^2)$  in normal linear regression is

$$\beta \sim MVN(\beta_0, \Sigma_0)$$

$$\sigma^2 \sim \text{Inverse-gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right).$$

# Semi-conjugate prior

## Theorem (Semi-conjugate prior)

*For normal linear regression, with previously defined semi-conjugate prior, it can be shown that*

$$\beta|y, X, \sigma^2 \sim MVN(\beta_n, V_{\beta_n})$$

$$\sigma^2|y, X, \beta \sim \text{Inverse} - \text{Gamma}\left(\frac{\nu_0 + n}{2}, \frac{\nu_0\sigma_0^2 + SSR(\beta)}{2}\right)$$

where  $\beta_n = (\Sigma_0^{-1} + \frac{X^T X}{\sigma^2})^{-1}(\Sigma_0^{-1}\beta_0 + \frac{X^T y}{\sigma^2})$ ,  $V_{\beta_n} = (\Sigma_0^{-1} + \frac{X^T X}{\sigma^2})^{-1}$ , and  $SSR(\beta) = (y - X\beta)^T (y - X\beta)$ .

Proof.



# Unit informative prior

To establish previous semi-conjugate prior, we need to specify  $(\beta_0, \Sigma_0, \nu_0, \sigma_0^2)$ . How can we make them informative for  $(\beta, \sigma^2)$ ?

We consider two weakly informative prior: Unit information prior and conjugate prior.

## Definition (Unit information prior)

If a prior distribution contains the same amount of information as that would be contained in only a single observation, then the prior distribution is called a unit information prior.

The unit informative prior for  $(\beta, \sigma^2)$  in LM is

$$\beta | \sigma^2 \sim MVN(\hat{\beta}_{OLS}, n\sigma^2(X^T X)^{-1})$$

$$\sigma^2 \sim \text{Inverse} - \text{Gamma}\left(\frac{1}{2}, \frac{\hat{\sigma}_{OLS}^2}{2}\right)$$

where  $\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$ , an estimate of  $\beta_0$  based on the marginal mean of  $y$ , and  $\hat{\sigma}_{OLS}^2 = \frac{SSR(\hat{\beta}_{OLS})}{n-p}$ .

Strictly speaking, not prior, as it requires knowledge of the observed data.  $\rightarrow$

Empirical Bayes method.

# Conjugate prior

The conjugate prior of  $(\beta, \sigma^2)$  is

$$\beta | \sigma^2 \sim \text{MVN}(0, g\sigma^2(X^T X)^{-1})$$

$$\sigma^2 \sim \text{Inverse} - \text{Gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right)$$

Above prior for  $\beta$  is a  $g$ -prior.

## Definition ( $g$ -prior)

In normal LM, the  $g$ -prior for  $\beta$  is a MVN with mean as a hyperparameter  $\beta_0$  and covariance matrix proportional to  $\sigma^2(X^T X)^{-1}$ , i.e.,

$$\beta | \sigma^2 \sim \text{MVN}(\beta_0, g\sigma^2(X^T X)^{-1})$$

## Theorem

*Posterior of  $\beta$  based on  $g$ -prior is invariant to changes in the scale of regressors.*

# Conjugate prior

With conjugate prior, we can show that

$$\beta|y, X, \sigma^2 \sim MVN\left(\frac{g}{g+1}(X^T X)^{-1}X^T y, \frac{g\sigma^2}{g+1}(X^T X)^{-1}\right)$$

$$\sigma^2|y, X \sim IG\left(\frac{\nu_0 + n}{2}, \frac{\nu_0\sigma_0^2 + y^T[I_n - \frac{g}{g+1}X(X^T X)^{-1}X]y}{2}\right)$$

Proof.





$$P(\sigma^2 | y, X) = \int \frac{P(y | X, \beta, \sigma^2)}{P(\beta | y, X, \sigma^2)} P(\beta | \sigma^2) \sim N \Rightarrow P(\sigma^2 | y, X)$$

$$\beta | y, X, \sigma^2 \sim MVN\left(\frac{g}{g+1}(X^T X)^{-1} X^T y, \frac{g\sigma^2}{g+1}(X^T X)^{-1}\right) \quad \sigma^2 | y$$

$$\beta | X, \sigma^2 \sim MVN(0, g\sigma^2(X^T X)^{-1})$$

$$\sigma^2 | X \sim IG\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right)$$

$$\frac{P(y, \beta | X, \sigma^2)}{P(y, \beta, X | \sigma^2)}$$

$$\Rightarrow P(\sigma^2 | X, \beta) \propto P(\beta | X, \sigma^2) P(\sigma^2 | X)$$

$$\begin{aligned} &= (2\pi)^{-\frac{n}{2}} \det(g\sigma^2(X^T X)^{-1})^{-\frac{1}{2}} \exp\left\{-\frac{1}{2} \beta^T (g\sigma^2(X^T X)^{-1})^{-1} \beta\right\} \cdot \\ &\quad \frac{1}{\Gamma(\frac{\nu_0}{2})} \left(\frac{\nu_0 \sigma_0^2}{2}\right)^{\frac{\nu_0}{2}} (\sigma^2)^{-\frac{\nu_0}{2}-1} \exp\left\{-\frac{1}{\sigma^2} \cdot \frac{\nu_0 \sigma_0^2}{2}\right\} \\ &\propto (\sigma^2)^{-(\frac{n+\nu_0}{2})-1} \exp\left\{-\frac{1}{\sigma^2} \left(\frac{\nu_0 \sigma_0^2}{2} + \frac{1}{2g}\right)\right\} \end{aligned}$$

$$\sim IG\left(\frac{n+\nu_0}{2}\right)$$

$$\Rightarrow \sigma^2 | X, \beta \sim I$$

$$y | X, \beta, \sigma^2 \sim MVN(X\beta, \sigma^2 I_n).$$

$$\beta \sim MVN(\beta_0, \Sigma_0)$$

$$\sigma^2 \sim \text{Inverse-gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right).$$

$$\beta | y, X, \sigma^2 \sim MVN(\beta_n, V_{\beta_n})$$

$$\sigma^2 | y, X, \beta \sim \text{Inverse-Gamma}\left(\frac{\nu_0 + n}{2}, \frac{\nu_0 \sigma_0^2 + SSR(\beta)}{2}\right)$$

where  $\beta_n = (\Sigma_0^{-1} + \frac{X^T X}{\sigma^2})^{-1} (\Sigma_0^{-1} \beta_0 + \frac{X^T y}{\sigma^2})$ ,  $V_{\beta_n} = (\Sigma_0^{-1} + \frac{X^T X}{\sigma^2})^{-1}$ , and  $SSR(\beta) = (y - X\beta)^T (y - X\beta)$ .

$$\sigma^2 | y, X \sim IG\left(\frac{\nu_0 + n}{2}, \frac{\nu_0 \sigma_0^2 + y^T [I_n - \frac{g}{g+1} X(X^T X)^{-1} X] y}{2}\right)$$

# Outline

- 1 Bayesian linear regression
- 2 Bayesian model selection

# Model selection in Bayesian linear regression

Choose variables in the following model through Bayesian ideas:

$$y_i = \sum_{j=1}^p \beta_j x_{ij} + \epsilon_i, \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2),$$

- Let  $\beta_j = z_j b_j$ , where  $z_j \in \{0, 1\}$ ,  $b_j$  is some real number, then

$$y_i = \sum_{j=1}^p z_j b_j x_{ij} + \epsilon_i,$$

- Each configuration of  $z = (z_1, \dots, z_p)^T$  corresponds to a specific model.
- Assume  $z_j$  to be random, the posterior of  $z_j$  for each  $j$  determines the final model

# Bayesian model selection

Question is how to derive the posterior distribution of  $z_j$ ?

- 1 Establish prior of  $b = (b_1, \dots, b_p)^T$
- 2 Establish prior of  $\sigma^2|z$
- 3 Derive  $P(y|X, z)$
- 4 Derive  $P(z_j|y, X, z_{-j})$  for  $j$ .

$$Y|X, \beta, \sigma^2 \sim \text{MVN}(X\beta, \sigma^2 I_n)$$

Conjugate prior:

$$\begin{cases} \beta|X, \sigma^2 \sim \text{MVN}(0, g\sigma^2(X^T X)^{-1}) \\ \sigma^2 \sim \text{IG}(\frac{v_0}{2}, \frac{v_0 \sigma_0^2}{2}) \end{cases}$$

$$\Rightarrow \begin{cases} \beta|y, X, \sigma^2 \sim \text{MVN}(\frac{g}{g+1}(X^T X)^{-1} X^T y, \frac{g\sigma^2}{g+1}(X^T X)^{-1}) \\ \sigma^2|y, X \sim \text{IG}(\frac{v_0+n}{2}, \frac{v_0 \sigma_0^2 + \text{SSR}_g(y)}{2}) \end{cases}$$

$$\text{with } \text{SSR}_g(y) = y^T (I_n - \frac{g}{g+1} X(X^T X)^{-1} X^T) y$$

Model selection:

$$Y_i = \sum_{j=1}^p \beta_j x_{ij} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid.}}{\sim} \mathcal{N}(0, \sigma^2)$$

$$\begin{cases} z = (z_1, \dots, z_p)^T & z_j \in \{0, 1\} \\ \beta_j = z_j b_j \end{cases}$$

$$\Rightarrow Y_i = \sum_{j=1}^p z_j b_j x_{ij} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid.}}{\sim} \mathcal{N}(0, \sigma^2)$$

根据  $z_j$  判断是否引入  $b_j$  ( $P(z_j=1|y, X) > 0.5$ ?)

$$P(z|Y, X) = \frac{P(Y|X, z) \pi(z)}{\sum_z P(Y|X, z) \pi(z)}$$

从 data generation 的角度理解

$$\textcircled{1} \begin{cases} z_j \in \pi(z_j), \quad P(z_j=1) = P(z_j=0) = \frac{1}{2} \\ b = (b_1, \dots, b_p)^T, \quad b|X, \sigma^2 \sim \text{MVN}(0, g\sigma^2(X^T X)^{-1}) \end{cases}$$

$$\Rightarrow \begin{cases} \beta|X, z, \sigma^2 \sim \text{MVN}(0, g\sigma^2(X_z^T X_z)^{-1}) \\ \sigma^2 \sim \text{IG}(\frac{v_0}{2}, \frac{v_0 \sigma_0^2}{2}) \end{cases}$$

② 从 data generation 的角度:

$$z \longrightarrow \beta_z \longrightarrow Y|X_z, \beta_z, \sigma^2 \sim \text{MVN}(X_z \beta_z, \sigma^2 I_n)$$

$\sigma^2 \nearrow$

# Gibbs sampling

For (s+1) iteration (given  $z^{(s)}, \beta^{(s)}, \sigma^{2(s)}$ )

① For each j,  $z_j^{(s+1)} \sim P(z_j | Y, X, \underline{z}^{(s)})$  ( $\underline{z}^{(s)} = (z_1^{(s)}, \dots, z_{j-1}^{(s)}, z_{j+1}^{(s)}, \dots, z_p^{(s)})$ )

$$P(z_j = 1 | Y, X, \underline{z}^{(s)}) = \frac{P(Y | X, z_j = 1, \underline{z}^{(s)})}{P(Y | X, z_j = 1, \underline{z}^{(s)}) + P(Y | X, z_j = 0, \underline{z}^{(s)})} \quad (\text{约分后})$$

其中  $P(Y | X, z) = \int_{\sigma^2} \int_{\beta} P(Y | X, z, \beta, \sigma^2) \cdot \pi(\beta | X, z, \sigma^2) \cdot \pi(\sigma^2) d\beta d\sigma^2$

②  $\sigma^{2(s+1)} | Y, X, z \sim IG(\frac{V_0 + n}{2}, \frac{V_0 \sigma_0^2 + SSR_g(Y)}{2})$

$$P(\sigma^2 | Y, X, z) = \frac{P(Y | X, z, \sigma^2) \cdot \pi(\sigma^2)}{\int_{\sigma^2} P(Y | X, z, \sigma^2) \cdot \pi(\sigma^2) d\sigma^2}$$

其中  $P(Y | X, z, \sigma^2) = \int_{\beta} P(Y | X, z, \beta, \sigma^2) \cdot \pi(\beta | X, z, \sigma^2) d\beta$

③ update  $\beta$ :

$$\beta^{(s+1)} | X, z^{(s+1)}, \sigma^{2(s+1)}, Y \sim MVN(\frac{g}{g+1} (X_z^T X_z)^{-1} X_z^T Y, \frac{g \sigma^{2(s+1)}}{g+1} (X_z^T X_z)^{-1})$$

If  $P(z_j = 1 | X, z) \approx \frac{1}{5} \sum_{s=1}^5 1(z^{(s)} = 1) > p_0$ , then  $x_j$  will be included

$$\hat{\beta} = \frac{1}{5} \sum_{s=1}^5 \beta^{(s)} \quad (\text{more robust})$$

Ex:  $\begin{cases} Y_{ij} = \alpha_i + \mu + \varepsilon_{ij} \\ \alpha_i \text{ iid. } N(0, \sigma_a^2) \end{cases}, \begin{cases} \varepsilon_{ij} \text{ iid. } N(0, \sigma_e^2) \\ \alpha_i \perp \varepsilon_{ij} \text{ for } \forall i, j \end{cases}$

$$P(\alpha_i | y) = \frac{P(y | \alpha_i) \pi(\alpha_i)}{\int_{\alpha} P(y | \alpha) \pi(\alpha) d\alpha}$$

其中  $P(y | \alpha_i) = \int_{\alpha_1} \int_{\alpha_2} \dots \int_{\alpha_{i-1}} \int_{\alpha_{i+1}} \dots \int_{\alpha_I} P(y | \alpha_1, \dots, \alpha_I) \cdot \pi(\alpha_1) \dots \pi(\alpha_{i-1}) \pi(\alpha_{i+1}) \dots \pi(\alpha_I) d\alpha_1 \dots d\alpha_I$

# Bayesian model averaged estimate of $\beta$

Estimate posterior mean of  $\beta$  by combining multiple models,

$$\hat{\beta}_{bma} = \frac{1}{S} \sum_{s=1}^S \beta^{(s)}$$

where  $\{\beta^{(1)}, \beta^{(2)}, \dots, \beta^{(S)}\}$  are sampled from  $P(\beta|y, X, z, \sigma^2)$ .

In real application of Bayesian LM,

- If no model selection required, get posterior of  $\beta$  and  $\sigma^2$  using analytical form
- If model selection required, consider Gibbs sampling to select model and using averaged estimator of  $\beta$  to obtain  $\hat{\beta}_{bma}$ .

## Metropolis

考虑以下情况:

$$p(y|\theta), \theta \sim \pi(\theta)$$

$$p(\theta|y) = \frac{p(y|\theta) \cdot \pi(\theta)}{\int_{\theta} p(y|\theta) \cdot \pi(\theta) d\theta}$$

其中  $\int_{\theta} p(y|\theta) \cdot \pi(\theta) d\theta$  可能无解析解 (没法 sample from)

解决思路:  $\{\theta^{(1)}, \dots, \theta^{(S)}, \theta^{(S+1)}, \dots, \theta^{(S')}\}$

· Markov chain:  $\theta^{(S)}$  relies on  $\theta^{(S-1)}$

· when  $S$  is large enough,  $\theta^{(S)} \sim p(\theta|y)$

Suppose we already have  $\{\theta^{(1)}, \dots, \theta^{(S')}\}$

$\theta^*$  based on  $\{\theta^{(1)}, \dots, \theta^{(S')}\}$

determine whether or not accept  $\theta^*$

$\theta^* \sim q(\theta|\theta^{(S)}) \Rightarrow$  proposal distribution

· If  $\frac{p(\theta^*|y)}{p(\theta^{(S)}|y)} > 1$ ,  $\theta^*$  is more likely to come from  $p(\theta|y)$  compare to  $\theta^{(S)}$ .

$\Rightarrow \theta^{(S+1)} = \theta^*$

$\Rightarrow \{\theta^{(1)}, \dots, \theta^{(S)}, \theta^{(S+1)}\}$

· If  $\gamma = \frac{p(\theta^*|y)}{p(\theta^{(S)}|y)} < 1$ , then accept  $\theta^*$  w.p.  $\gamma$

Implementation:

$u \sim \text{Unif}(0, 1)$

$\begin{cases} \text{if } u \leq \gamma, & \theta^{(S+1)} = \theta^* \\ \text{if } u > \gamma, & \theta^{(S+1)} = \theta^{(S)} \end{cases}$

接下来讨论 proposal distribution  $q(\theta|\theta^{(S)})$  的选取.

在 Metropolis algorithm 中,  $q(\theta|\theta^{(S)})$  为 symmetric distribution:

$$q(\theta^*|\theta^{(S)}) = q(\theta^{(S)}|\theta^*)$$



e.g.  $\text{Unif}(\theta^{(s)} - \delta, \theta^{(s)} + \delta)$   
 $N(\theta^{(s)}, \sigma^2)$

Metropolis algorithm:

• Metropolis algorithm.  
 Given  $\{\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(s)}\}$ .  
 ① Sample  $\theta^* \sim q(\theta|\theta^{(s)})$ ,  $q(\theta|\theta^{(s)})$  is a symmetric proposal dist.  
 ② Calculate  $r = \min\left\{\frac{p(\theta^*|y)}{p(\theta^{(s)}|y)}, 1\right\}$ .  
 ③ If  $r=1$ ,  $\theta^{(s+1)} = \theta^*$ .  
 If  $r < 1$ , sample  $u \sim U(0,1)$ . then  
 $\theta^{(s+1)} = \begin{cases} \theta^*, & \text{if } u \leq r \\ \theta^{(s)}, & \text{if } u > r. \end{cases}$

Ex:  $\begin{cases} Y_i | X_i, \beta \sim \text{Pois}(e^{X_i^T \beta}) \\ \beta \sim N(0, \Sigma) \end{cases}$

Given  $\{\beta^{(1)} \dots \beta^{(s)}\}$

$$\textcircled{1} \beta^* | \beta^{(s)} \sim \text{MVN}(\beta^{(s)}, \sigma^2 (X^T X)^{-1})$$

↪  $\text{var}(\log(\gamma + 0.5))$  可理解为  $g$ -prior 中的  $g$  的选取

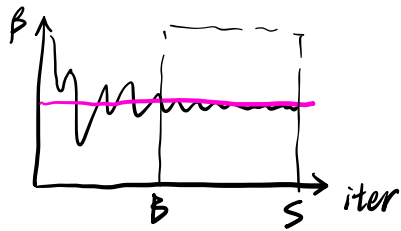
$$\textcircled{2} \frac{p(\theta^*|y)}{p(\theta^{(s)}|y)} = \frac{p(y|\beta^*) \pi(\beta^*)}{p(y|\beta^{(s)}) \pi(\beta^{(s)})}$$

$$\begin{aligned} p(y|\beta^*) &= p(Y_1, \dots, Y_n | \beta^*) \\ &= \prod_{i=1}^n p(Y_i | \beta^*) \\ &= \prod_{i=1}^n \frac{(e^{X_i^T \beta^*})^{Y_i}}{Y_i!} \cdot \exp\{-e^{X_i^T \beta^*}\} \end{aligned}$$

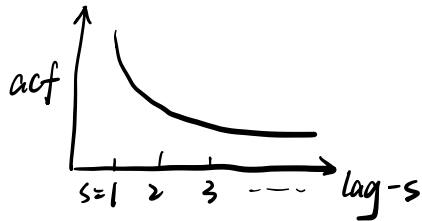
$$\textcircled{3} \text{ If } r = \min\left\{\frac{p(\beta^*|y)}{p(\beta^{(s)}|y)}, 1\right\} = 1, \beta^{(s+1)} = \beta^*$$

$$\text{If } r < 1, \beta^{(s+1)} = \begin{cases} \beta^* & \text{if } r \geq u \sim \text{Unif}(0,1) \\ \beta^{(s)} & \text{if } r < u \end{cases}$$

判断收敛:



还可以作出 Auto-correlation plot



$\{X_1, \dots, X_n\}$

lag-1 correlation:  $\tilde{X} = \{X_1, \dots, X_{n-1}\}$   $\Rightarrow$  算 Pearson correlation  
 $\tilde{Y} = \{X_2, \dots, X_n\}$

lag-2 correlation:  $\tilde{X} = \{X_1, \dots, X_{n-2}\}$   $\Rightarrow$  算 Pearson correlation  
 $\tilde{Y} = \{X_3, \dots, X_n\}$